

Conditionals

Three parts. Do Part 1 on paper before you touch a computer. Parts 2 and 3 go in jGRASP.

Name _____ Date _____

Every file starts with a comment giving your name, the date, and one sentence on what the program does. Use only what you have been taught: if / else if / else, Scanner, printf and casting. No loops, no switch, and no methods other than main.

Part 1

Predict the output

Write down what each one prints. No running it first, that is the whole point. Then run them and see how you did.

1.

```
int hp = 12;
if (hp > 10) {
    System.out.print("Healthy ");
}
System.out.print("Done");
```

2.

```
int hp = 3;
if (hp > 10) {
    System.out.print("Healthy ");
} else {
    System.out.print("Hurt ");
}
System.out.print("Done");
```

3.

```
int coins = 50;
if (coins >= 10) {
    System.out.println("Bronze");
} else if (coins >= 100) {
    System.out.println("Silver");
} else if (coins >= 500) {
    System.out.println("Gold");
}
```

```
}
```

4.

```
int lives = 3;
if (lives == 0);
{
    System.out.println("Game over");
}
```

5.

```
int x = 7;
if (x > 5 && x < 10) {
    System.out.println("In range");
}
if (x < 5 || x > 6) {
    System.out.println("Outside the middle");
}
```

6.

Go back to number 3. What would you have to change so that 500 coins prints Gold? Write the fix.

Part 2

Fix the bug

Each of these compiles without a single error. Each one is wrong. Find the bug, fix it, and write one sentence explaining what was happening.

Bug 1

```
Scanner input = new Scanner(System.in);
System.out.print("Password: ");
String password = input.next();

if (password == "dragon") {
    System.out.println("Access granted");
} else {
    System.out.println("Access denied");
}
```

It always denies you, even when you type dragon.

Bug 2

```
int score = 95;
String grade;

if (score >= 60) {
    grade = "D";
} else if (score >= 70) {
    grade = "C";
} else if (score >= 80) {
    grade = "B";
} else if (score >= 90) {
    grade = "A";
} else {
    grade = "F";
}

System.out.println(grade);
```

A 95 should be an A. It prints D.

Bug 3

```
int hp = 100;

if (hp <= 0)
    System.out.println("You died.");
    System.out.println("Respawning...");
```

It says Respawning... even at full health.

Part 3

Write these

A. Level checker

Ask for XP, print the level.

XP	Level
1000 or more	4
500 to 999	3
100 to 499	2
under 100	1

```
Enter your XP: 620
Level 3
```

Use one if / else if / else chain. Not four separate if statements.

B. Magic 8-Ball

Ask the user a yes-or-no question, ignore what they typed, and print one of at least five random answers.

```
Ask me anything: Will I pass the test?  
Signs point to yes.
```

You will need `(int) (Math.random() * 5)` to pick a number, then a chain to turn that number into an answer.

C. One-shot guessing game

This is the main one.

- Ask the player's name
- Pick a secret number from 1 to 10 at random
- Ask for one guess
- Tell them **too high**, **too low**, or **exactly right**
- If they were off by exactly 1 in either direction, also tell them they were close
- Always reveal the answer at the end

```
Your name: Sam  
Guess 1-10, Sam: 7  
  
Too high!  
But you were close.  
It was 6.
```

Reading a name and then a number in the same program is where `nextLine()` gets weird. `nextInt()` takes the number but leaves the Enter key sitting in the buffer, so the next `nextLine()` reads that leftover instead of waiting for you. Order your reads carefully.

D. Text adventure

Write a short dungeon scene that gives the player a real choice. One door on the left, one on the right, and they lead to different rooms with different text. Use `.equals()` or `equalsIgnoreCase()` to check what they typed, never `==`.

Add at least one thing that changes their HP or gold, then print their status afterward.

Challenge, optional

Leap year

Ask for a year and say whether it is a leap year. The rule is trickier than it looks:

- Divisible by 4, leap year
- Unless it is also divisible by 100, then not a leap year
- Unless it is *also* divisible by 400, then leap year after all

So 2024 is, 1900 is not, and 2000 is. Use % and either a nested if or a single condition with && and ||. Both work. The one-line version is very satisfying once it clicks.

Rules for all of it

1. **Always use braces**, even when the body is one line.
2. **No semicolon** after the closing) of an if.
3. Indent everything inside braces. Java does not care. Whoever reads it does.
4. If you get stuck for more than fifteen minutes on one thing, stop and ask for help.

Tick these off before you hand it in

- ☐ Part 1 answers written on paper, before running anything
- ☐ All three bugs fixed, each with one sentence explaining it
- ☐ Level checker uses one chain, not four separate ifs
- ☐ Magic 8-Ball has at least five answers
- ☐ Guessing game handles high, low, exact, and close
- ☐ Dungeon has two doors that lead somewhere different
- ☐ Braces on every if, even the one-liners
- ☐ Name, date and description at the top of every file

Battle Sim

Two fighters face off. Your program asks about each one, works out a power score for each, and says who wins.

File name BattleSim.java • **Practice with** if / else if / else, Scanner, casting, printf

Use only what you have learned so far. No loops, and no methods other than `main`.

Each fighter gets two subscores between 0 and 100. Add them for a total between 0 and 200. Highest total wins.

Step 1

The combat score

Ask whether the fighter is a **1) Warrior** or a **2) Mage**, then ask for their stats.

If Warrior, ask for strength, armor and stamina, each 1 to 100:

```
combat = (2 * strength + armor + stamina) / 4.0
```

If Mage, ask for intellect, mana and focus, each 1 to 100:

```
combat = (intellect + 2 * mana + focus) / 4.0
```

Notice the `4.0` in both. Think about why it is not `4`.

Step 2

The gear score

Ask for three things:

- the fighter's level, a whole number
- the max level in the game, a whole number
- a gear multiplier between 0.5 and 1.0, a decimal

```
gear = level / maxLevel * 100 * gearMultiplier
```

Careful. Level and max level are both whole numbers. Read that formula and work out what Java is going to do with them before you write it.

Step 3

Compare

Total score is `combat + gear`. Do this for both fighters, then print exactly one of these:

```
Fighter 1 wins!  
Fighter 2 wins!  
It's a draw!
```

Sample run

What you type is in bold.

```
BATTLE SIM  
Enter two fighters and I'll tell you who wins.  
  
Fighter 1:  
  1) Warrior or 2) Mage? 1  
  Strength? 80  
  Armor? 60  
  Stamina? 55  
  combat score = 68.8  
  Level? 34  
  Max level? 50  
  Gear multiplier? 0.9  
  gear score = 61.2  
  
Fighter 2:  
  1) Warrior or 2) Mage? 2  
  Intellect? 70  
  Mana? 85  
  Focus? 60  
  combat score = 75.0  
  Level? 28  
  Max level? 50  
  Gear multiplier? 0.95  
  gear score = 53.2  
  
Fighter 1 total = 130.0  
Fighter 2 total = 128.2  
  
Fighter 1 wins!
```

Printing one decimal place

Your scores will come out looking like `68.7500000001`. To print just one decimal:

```
System.out.printf("  combat score = %.1f\n", combat);
```

The `%.1f` is a placeholder that gets replaced by the number, rounded to one decimal for display only. The `\n` moves to the next line, since `printf` does not do that on its own.

Only the *display* rounds. Compare the real, unrounded totals when you decide the winner. That means the printed numbers might look like they do not quite add up. That is correct and expected.

Requirements

- ☐ Reads all input with one `Scanner` object, made once at the top
- ☐ Uses `int` and `nextInt()` for whole numbers, `double` and `nextDouble()` for decimals
- ☐ Branches correctly on Warrior vs Mage for both fighters
- ☐ Every score prints to exactly one decimal place
- ☐ Prints exactly one of the three result lines
- ☐ Braces on every `if`, even one-liners
- ☐ A comment at the top with your name, the date, and one sentence on what the program does
- ☐ Output matches the sample layout, including the indenting

Stretch goals

Only after the above works.

1. **Class advantage.** Warriors beat Mages in a tie. Add that as an extra condition on the draw case.
2. **Rank the winner.** Print a rank based on the winning total: 180 or more is Legendary, 150 or more is Elite, 100 or more is Seasoned, below that is Rookie.
3. **A third class.** Add 3) Rogue with its own three stats and its own formula. You decide what makes a Rogue strong.
4. **Crit chance.** Give the winner a 1-in-4 chance of landing a critical hit, and print an extra line when they do. You will need `Math.random()`.

Two things that will get you

The gear formula

`level / maxLevel` with two whole numbers is integer division. 34 divided by 50 is 0 in Java, so every gear score comes out as 0.0. Fix it the same way you fix any division that should not have been whole-number division.

Reading a number after nextInt()

If you add any text input to this program, the leftover Enter key is waiting for you. `nextInt()` reads the digits and stops, leaving the newline behind, so the next `nextLine()` comes back empty.

Battle Sim is adapted from UW CSE 142 assignment 4, Admissions. Starter files and the online version of this assignment are on the course site.